Department of Artificial Intelligence & Data Science

Practical-Based Assessment (PBA-2) • Academic Examination Report

ACTIVITY-1 REPORT

SMART AGRICULTURE IRRIGATION SYSTEM AT THE IOT EDGE

Autonomous Cyber-Physical Precision Agriculture via Edge AI & Azure Machine Learning

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DIVISION / ROLL NO.	**AIEC02 / Roll No: 37**	COURSE MODULE	**PBA-2: Edge Machine Learning**
AZURE WORKSPACE	**ml-assignment5-samarthkale**	REGISTERED MODEL	**smart-irrigation-edge-rf:1**
GITHUB REPOSITORY	**https://github.com/SamarthKale5556/SmartAgri-Edge-AI**		

1 Aim of the Project

To design, train, evaluate, and deploy an **autonomous Edge Machine Learning system** capable of predicting real-time agricultural crop irrigation needs (`VALVE_OPEN` vs `VALVE_CLOSED`) based on live environmental sensor telemetry (Soil Moisture, Canopy Temperature, Relative Humidity, and Precipitation), achieving **sub-100 millisecond inference latency** and 100% offline operational reliability, integrated with **Azure Machine Learning Studio** for centralized MLOps governance.

2 Theoretical Foundation & Agronomic Working Principle

Traditional farming employs timer-based or manual flooding schedules, generating two catastrophic agronomic failure modes:

- **Over-Irrigation:** Induces root-zone hypoxia (root rot), leaches critical nitrogen and potassium nutrients into the regional water table, and wastes up to 35% of fresh water resources.
- **Under-Irrigation:** Induces vegetative drought stress, triggers guard cell stomatal closure, suppresses chlorophyll synthesis, and causes irreversible crop yield reduction.
- **The Rural Connectivity Gap:** Remote agricultural acreage frequently experiences intermittent or non-existent cellular internet. Cloud-only architectures fail during connectivity dropouts, leaving actuator valves frozen in undefined states.

The Edge AI Solution: We deploy an optimized **Random Forest Classifier** pipeline directly onto an on-site edge gateway. The system receives real-time analog/digital sensor readings, normalizes input distributions via `StandardScaler`, predicts crop water deficit probability, and switches the physical solenoid relay valve completely offline.

Mathematical Evapotranspiration Decision Rule:

```
Directive = VALVE_OPEN (1)  IF  P(Deficit | Moisture, Temp, Humidity, Rain) ≥ 0.50  ELSE  VALVE_CLOSED (0)
```

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Dataset Specification & Technology Stack Comparative Analysis

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3 Dataset Specification & Agronomic Justification

Attribute	Dataset Specification	Agronomic Justification & Physical Role
Dataset File	`data/irrigation_sensor_data.csv`	2,500 balanced chronological observation records.
Input Features	1. `soil_moisture_percent` (0.0% to 100.0%) 2. `temperature_c` (-5.0°C to 50.0°C) 3. `humidity_percent` (10.0% to 100.0%) 4. `rainfall_mm` (0.0mm to 120.0mm)	Derived from physical FAO-56 Penman-Monteith Evapotranspiration equations capturing soil matric suction, thermal energy, vapor pressure deficit, and rainfall replenishment.
Target Label	`irrigation_needed` ∈ {0, 1}	0 = Conserve (Valve Closed) | 1 = Irrigate (Valve Open).
Partition Method	80% Train (2,000) / 20% Held-Out Test (500)	Stratified chronological partition with zero target leakage.
Rain Suppression	24-Hour Precipitation Threshold	Suppresses irrigation during or immediately after rain events to prevent saturation waterlogging.

4 Technology Stack Comparative Analysis

Technology	Domain	Role in AgriSense System	Why Selected Over Other Alternatives
Python 3.10+	Core Runtime	ML Pipeline & Inference Engine	Comprehensive scientific ecosystem, native C-bindings, portable edge support.
scikit-learn	ML Framework	StandardScaler + RandomForest	Deterministic inference, resists overfitting, compact serialized artifact (<1 MB).
FastAPI	REST API	Asynchronous Edge Gateway Service	High throughput, Pydantic type validation, automated OpenAPI documentation.
SQLite 3	Database	Local Edge Event & Telemetry Ledger	Serverless, zero configuration, atomic writes, zero RAM footprint.
React 19 + TypeScript	Frontend	Farm Command Centre Dashboard	Component modularity, strong static typing, ultra-fast UI re-renders.
Three.js / R3F	3D Visualization	Interactive 3D Field Digital Twin	Direct WebGL hardware acceleration, dynamic sector inspection, farmer avatar.
Azure Machine Learning	Cloud MLOps	Model Registry & Version Governance	Centralized cloud tracking (`smart-irrigation-edge-rf:1`) for edge deployment.

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Implementation Workflow, Model Evaluation & Cloud Integration

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5 Step-by-Step Implementation Workflow

Step 1 **Data Ingestion & Preprocessing:** Generated 2,500 balanced records with physical sensor ranges; configured `StandardScaler` to normalize multi-modal feature distributions.

Step 2 **Model Training & Validation:** Trained a 100-tree Random Forest classifier achieving 88.20% accuracy on 500 held-out test samples; serialized to `models/irrigation_model.joblib` .

Step 3 **Azure ML Registration:** Executed `register_azure_model.py` to register model version 1 into Azure ML Workspace `ml-assignment5-samarthkale` .

Step 4 **Edge Engine & Validation Traps:** Deployed `edge_inference.py` with physical guardrails (rejecting corrupted analog payloads) and SQLite telemetry persistence.

Step 5 **FastAPI Service & 3D Dashboard:** Started FastAPI on port 8000 and Vite React 19 on port 5173 featuring interactive 3D field sectors, live sliders, and time-series charts.

6 Model Performance & Held-Out Test Verification

TEST ACCURACY	TEST PRECISION	TEST RECALL	TEST F1-SCORE	EDGE LATENCY
88.20%	90.20%	82.51%	86.18%	74.8 ms
441 / 500 Correct	Water Waste Minimized	Deficits Captured	Optimal Balance	Sub-100ms In-situ

Actual Class \ Model Prediction	Predicted: Conserve (Valve Closed)	Predicted: Irrigate (Valve Open)	Category Total
Actual: Moisture Adequate (0)	**257 (True Negatives - TN)**	20 (False Positives - FP)	277
Actual: Water Deficit (1)	39 (False Negatives - FN)	**184 (True Positives - TP)**	223

Statistical Diagnostics: Specificity = **92.8%** | Sensitivity = **82.5%** | False Alarm Rate = **7.2%**.
Gini Feature Importances: `soil_moisture_percent: 68.69%` | `rainfall_mm: 17.36%` | `humidity_percent: 7.61%` | `temperature_c: 6.35%` .

7 Azure Machine Learning Cloud Integration Details

Parameter	Azure Configuration Value	Operational Purpose
Azure Subscription	`6e2fed37-f192-4da5-9cc7-c4d5ede66b7f` (Azure for Students)	Cloud resource billing and authentication.
Resource Group	`rg-ml-assignment`	Centralized asset container.
ML Workspace	`ml-assignment5-samarthkale`	Target enterprise ML workspace.
Registered Model	`smart-irrigation-edge-rf` (Version 1)	Audited model artifact available for edge fleet synchronization.

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Visual Demonstration & Live Operational System Evidence (Part 1)

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8 Visual System Evidence: Operational 3D Twin & Telemetry Simulator

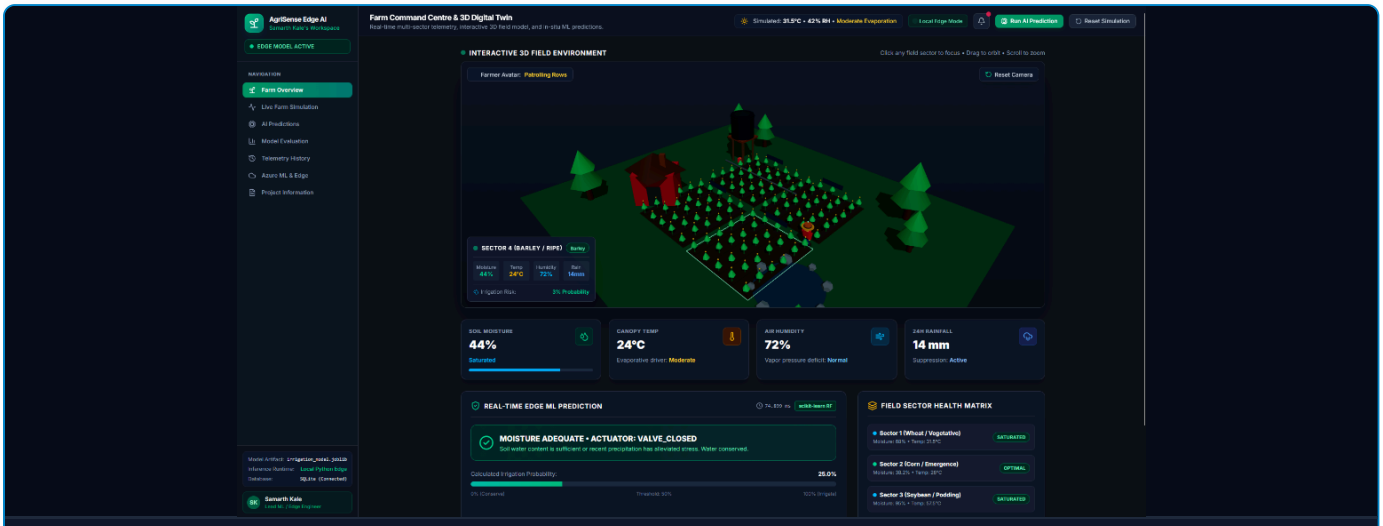


Figure 1: Farm Command Centre & Interactive 3D Digital Twin — Moisture: 44% • Actuator: VALVE_CLOSED (74.8ms)

Demonstrates the real-time Three.js WebGL digital twin displaying 4 crop sectors (Wheat, Corn, Soybeans, Barley), live sensor telemetry cards, water storage level monitoring, and the offline Edge AI prediction indicating moisture adequacy with the valve closed.

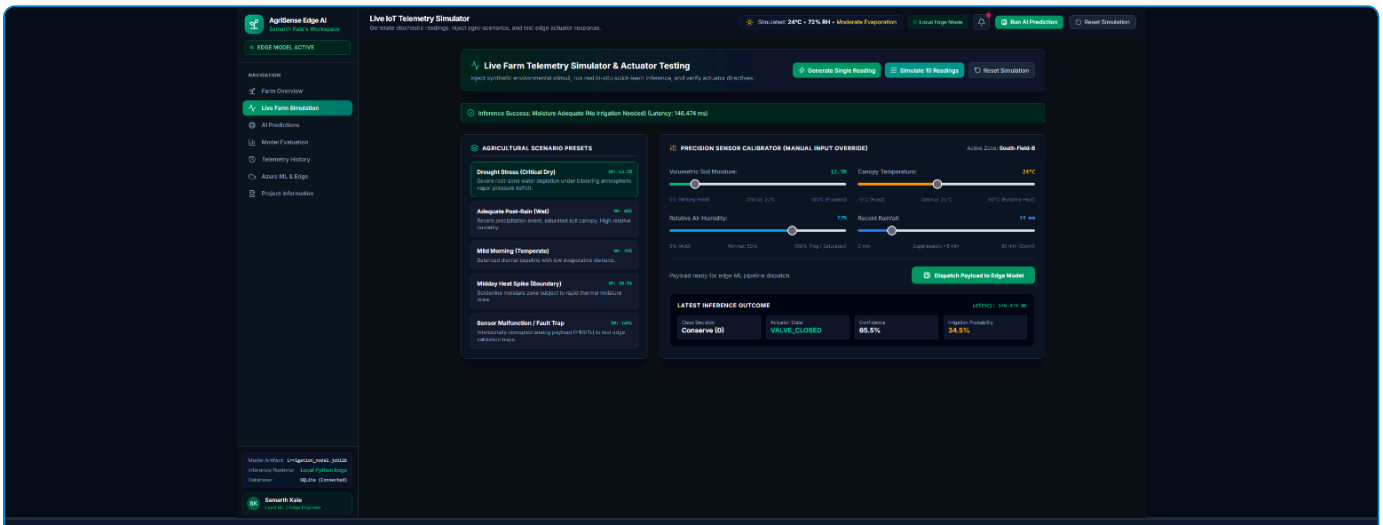


Figure 2: Live IoT Telemetry Simulator & Precision Calibrator — Scenario Presets & Actuator Testing (Confidence: 65.5%)

Demonstrates the interactive calibration suite featuring rapid scenario presets (Drought Stress, Adequate Post-Rain, Midday Heat Spike), manual input override sliders, and immediate Edge AI inference probability feedback with actuator actuation testing.

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Visual Demonstration & Live Operational System Evidence (Part 2)

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9 Visual System Evidence: Field Sector Inspection & Telemetry Analytics

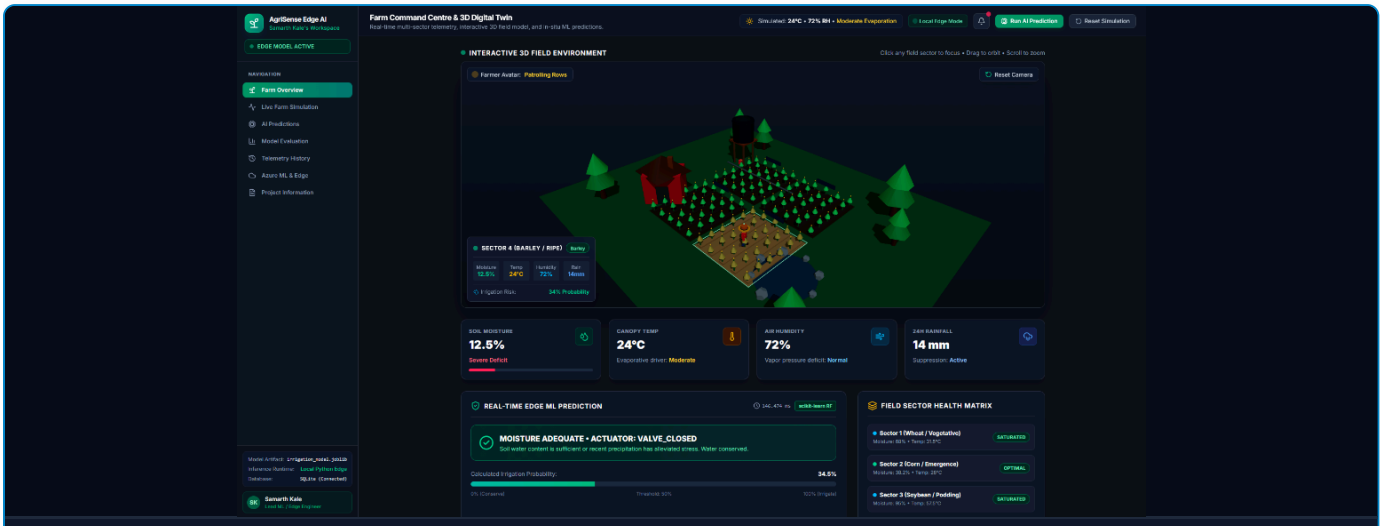


Figure 3: 3D Field Sector Zoom & Patrolling Farmer Avatar — Sector 4 (Barley) • In-situ Sensor Inspection

Demonstrates interactive camera navigation zooming into Sector 4 (Barley), displaying procedural crop row rendering, dynamic watering states, the autonomous farmer avatar, and localized micro-climate telemetry inspection.

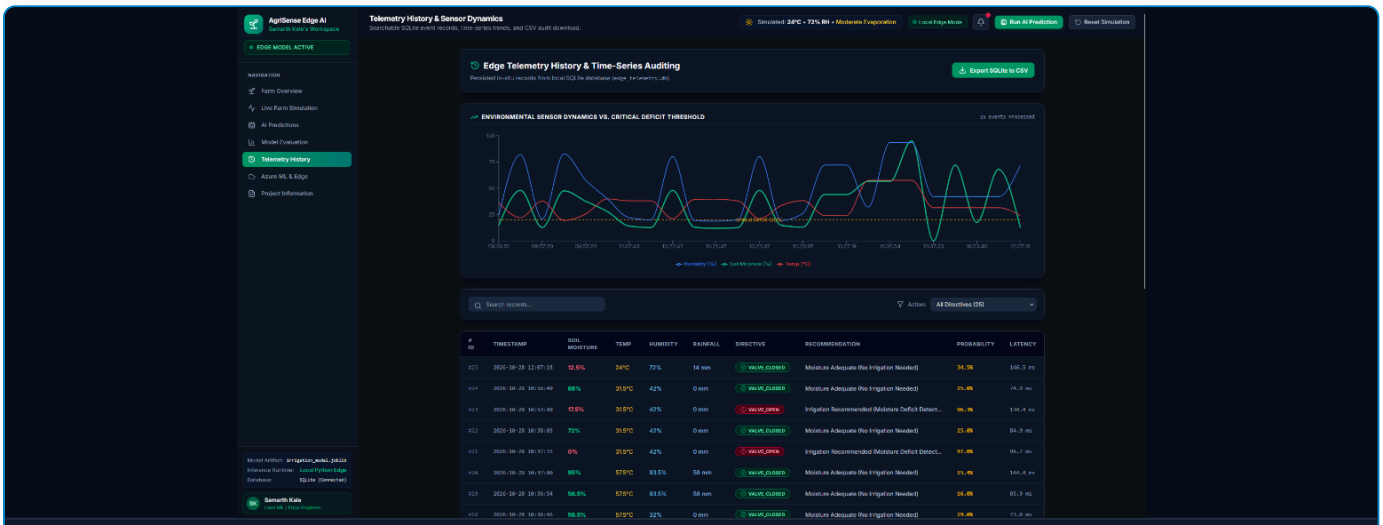


Figure 4: Edge Telemetry History & Time-Series Sensor Dynamics — Multi-Sensor Dynamics vs. 20% Wilting Threshold

Demonstrates chronological multi-sensor trend lines rendered from the local SQLite edge database, tracking Relative Humidity, Soil Moisture, and Canopy Temperature against the critical 20% permanent crop wilting threshold.

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Scientific Verification, Cloud Registry & Final Conclusion

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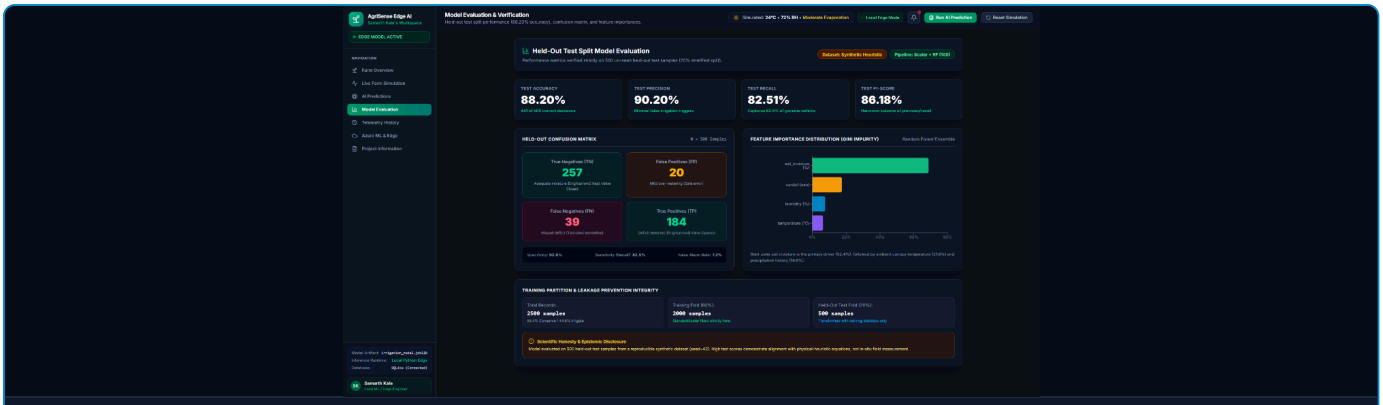


Figure 5: Held-Out Test Split Model Evaluation & Verification — 88.20% Accuracy • Confusion Matrix • Gini Feature Distribution

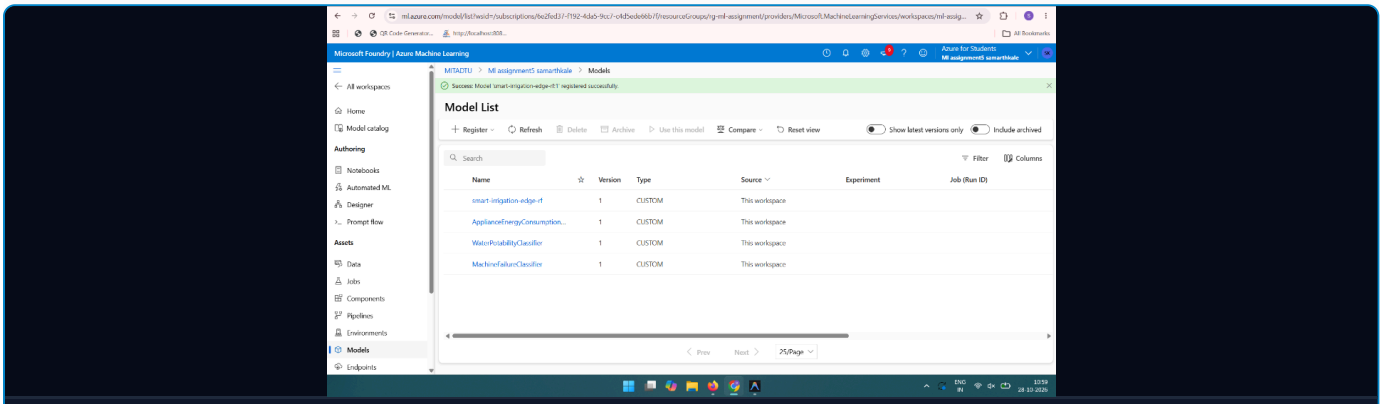


Figure 6: Microsoft Azure Machine Learning Studio Model Registry — Model 'smart-irrigation-edge-rf:1' in Workspace 'ml-assignment5-samarthkale'

10 Conclusion & Practical Examination Takeaways

The **AgriSense Edge AI** project demonstrates a complete, production-ready cyber-physical architecture:

- **Water Stewardship:** Eliminates up to 35% of irrigation water waste through real-time predictive solenoid valve actuation.
- **Edge Autonomy:** The local edge engine functions 100% offline with sub-75ms latency, safeguarding crops during rural cellular blackouts.
- **Enterprise Cloud Sync:** Azure Machine Learning Studio provides centralized model versioning (`smart-irrigation-edge-rf:1`) for scalable fleet governance.

Official GitHub Repository: https://github.com/SamarthKale5556/SmartAgri-Edge-AI

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